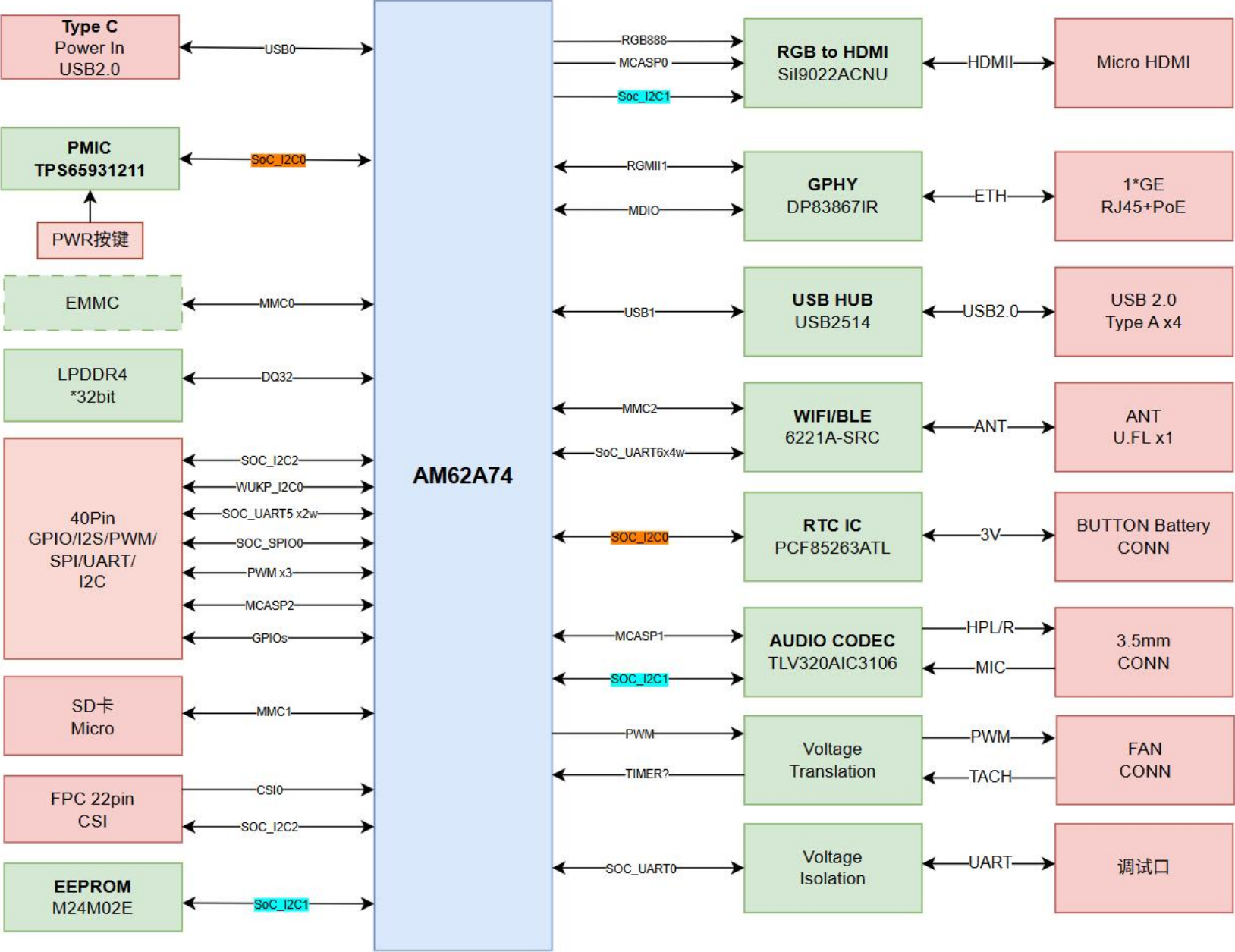
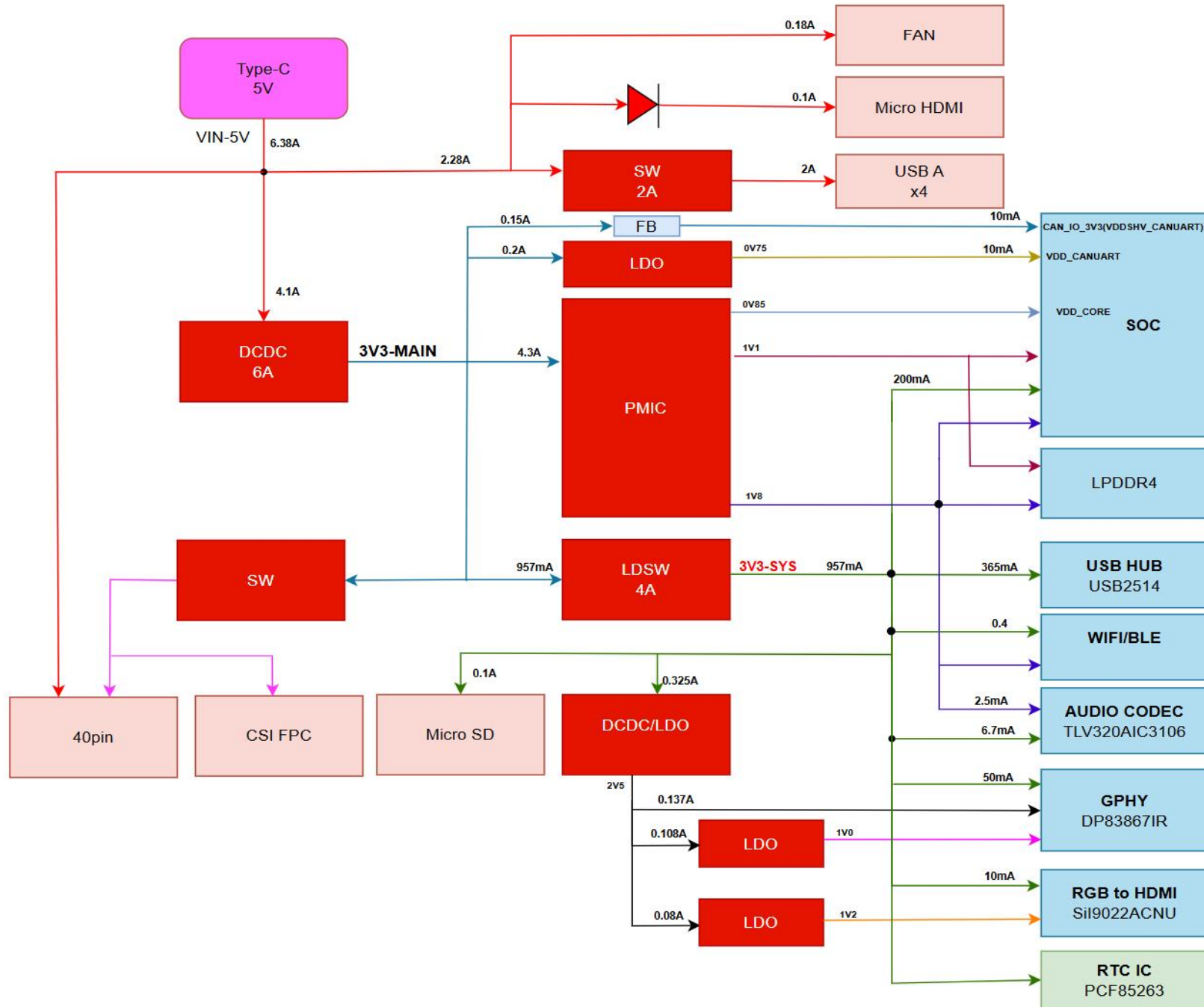


BLOCK DIAGRAM AM62A



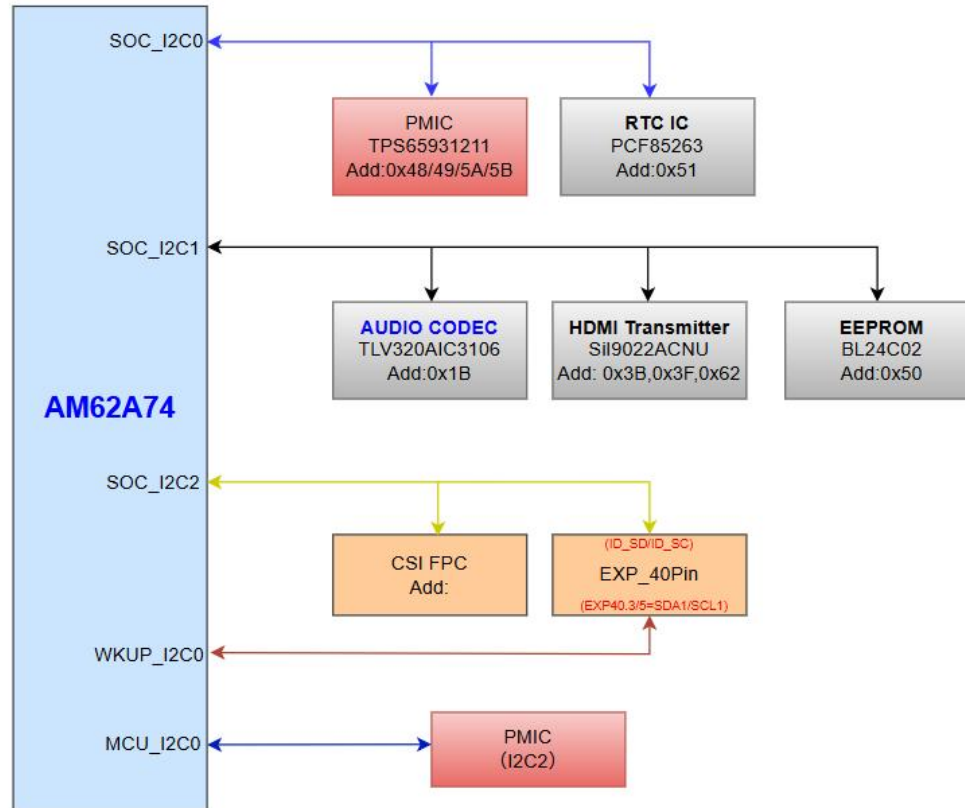
POWER BLOCK DGM



NOR

 Beijing InHand Networks Technology Co., Ltd.	
Project:	Mo_62A
File:	02. POWER BLOCK DGM
Date:	Wednesday, April 01, 2026
Designed by:	Reviewed by:
Rev:	V1.0
Sheet:	2 of 22

I2C TREE



NOR

 Beijing InHand Networks Technology Co., Ltd.					
Project:	Mo_62A				
File:	03. I2C TREE				
Date:	Wednesday, April 01, 2026	Rev:	V1.0		
Designed by:		Reviewed by:		Sheet:	3 of 22

SOC POWER SUPPLY PMIC

Cad Note :-
Tie the feedback after the
DC/DC output bulk caps or
use remote sensing

VDDA 1V8

Mount R121 & Demount R110
when VCC_CORE is 0.75V

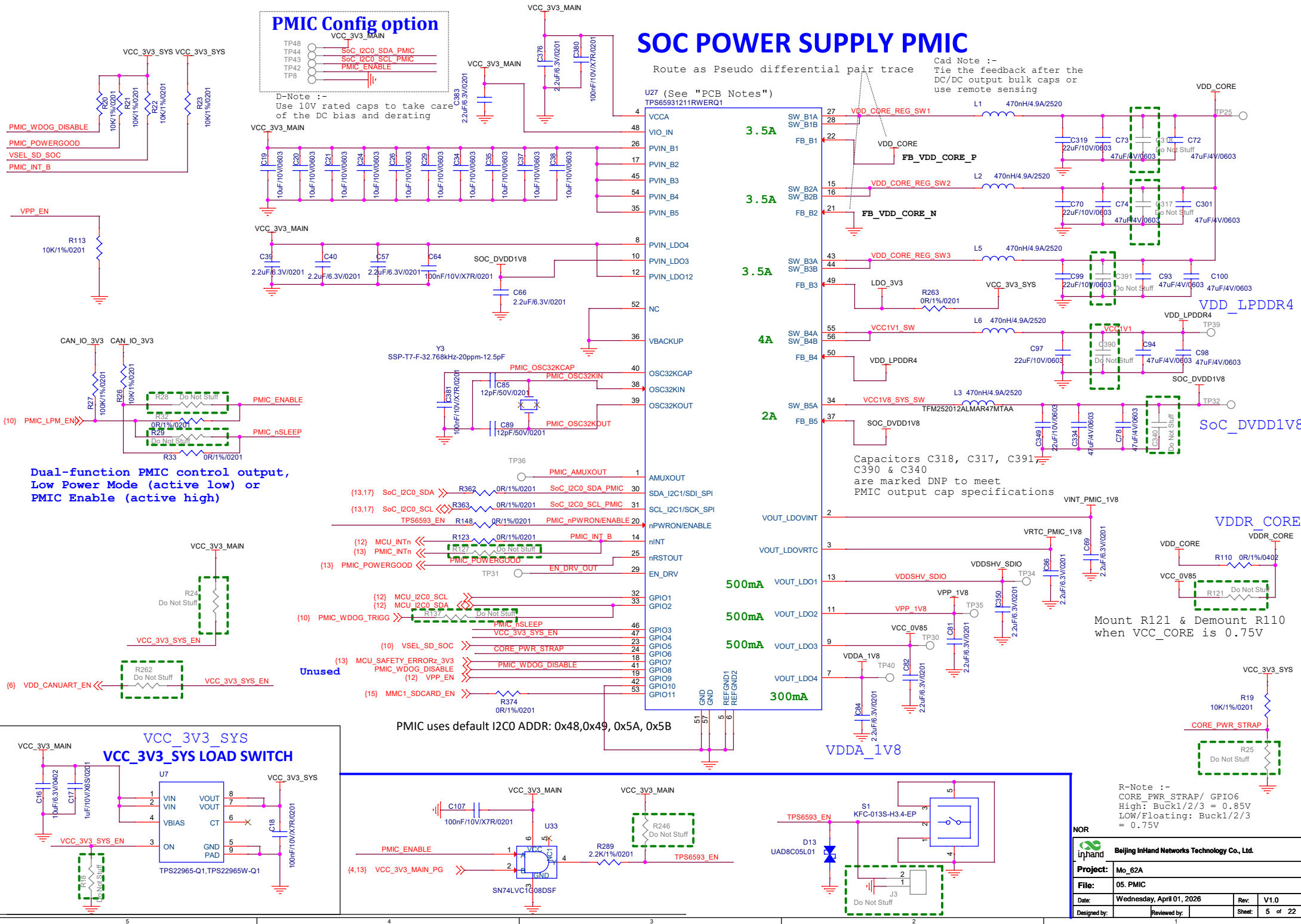
Project: Mo_62A

File:	05. PMIC
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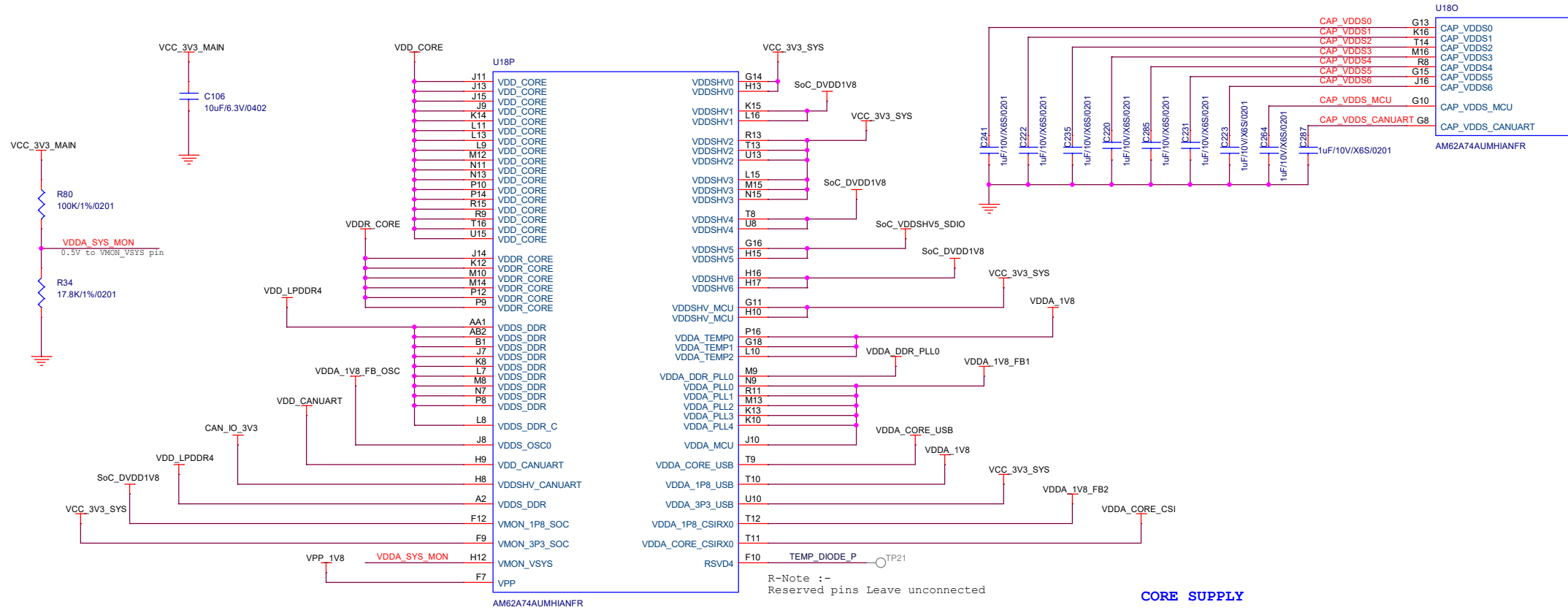
File:	00000000		
Date:	Wednesday, April 01, 2026	Page:	1410

Date:	Wednesday, April 01, 2020	Rev:	V1.0
Declaration:		Sheet:	5 of 22

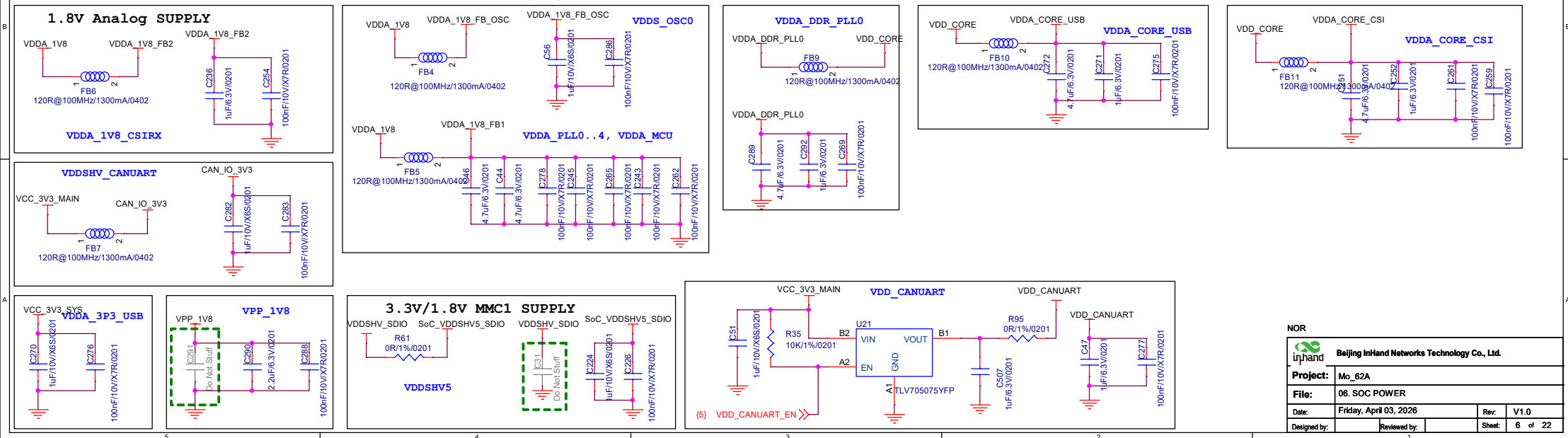
Designed by:	Reviewed by:	Sheet	3 of 22
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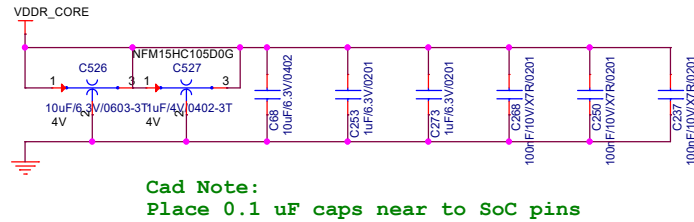
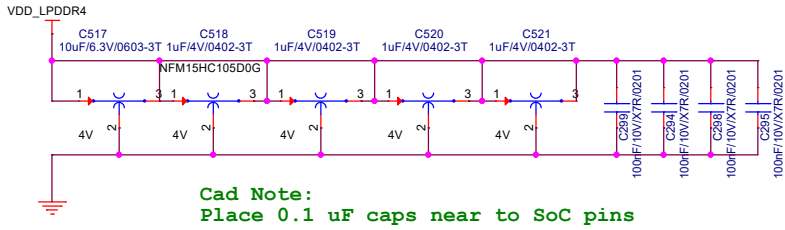
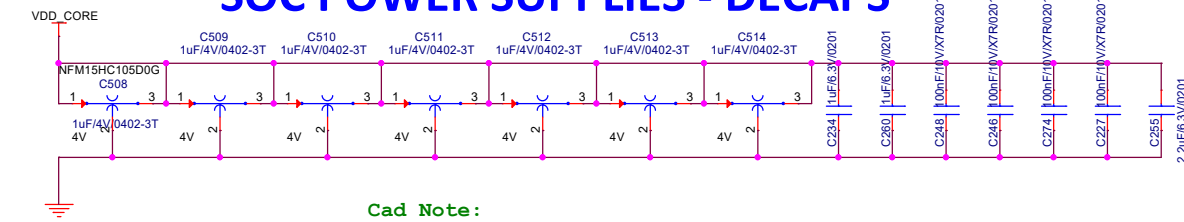
SOC POWER



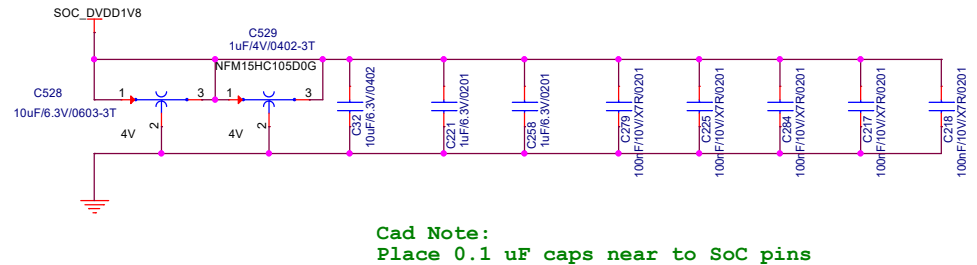
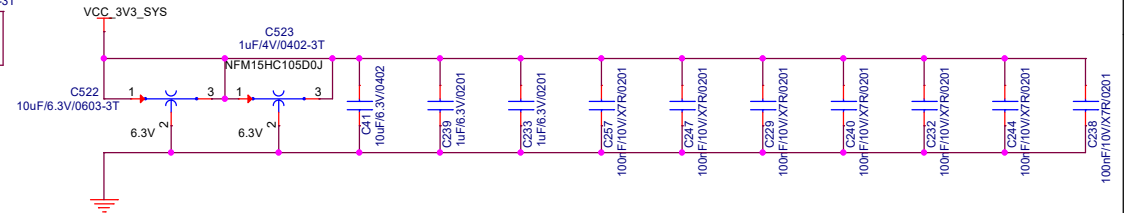
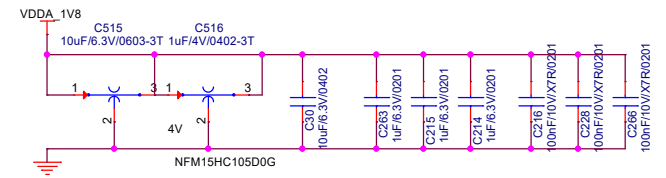
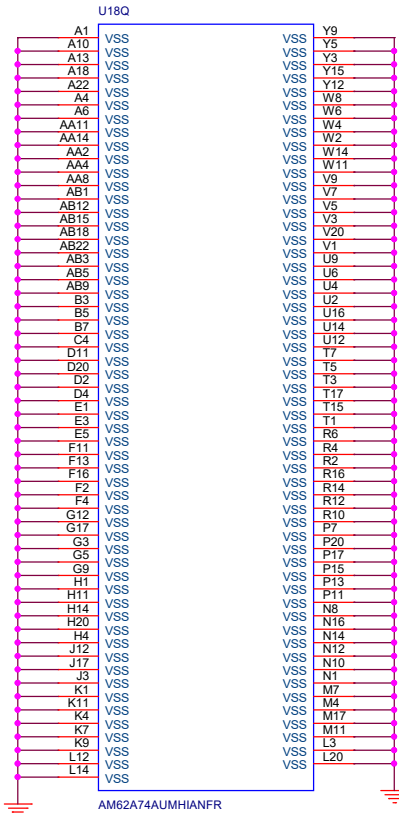
CORE SUPPLY



SOC POWER SUPPLIES - DECAPS



SOC VSS

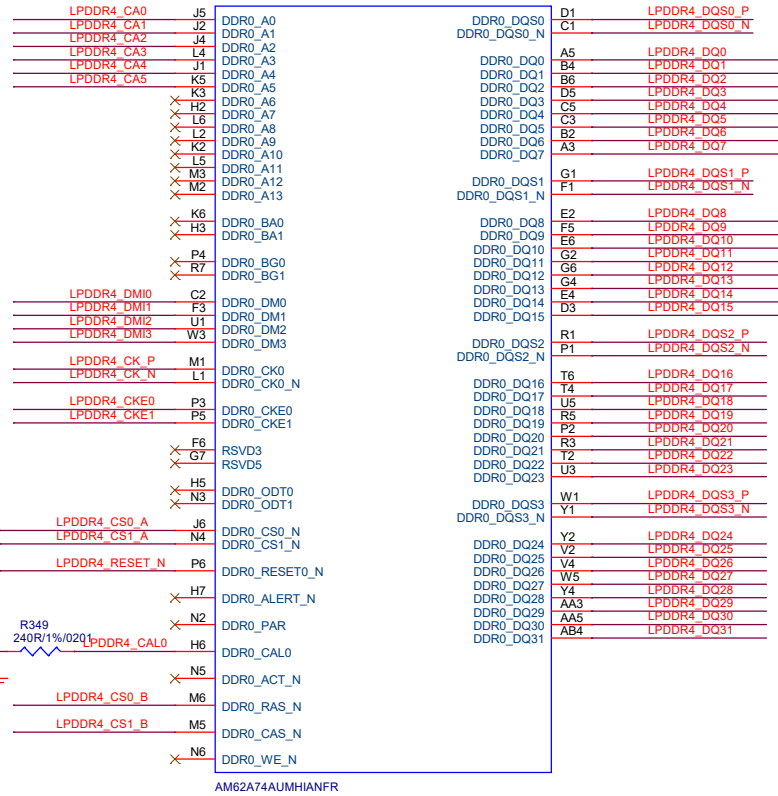


NOR

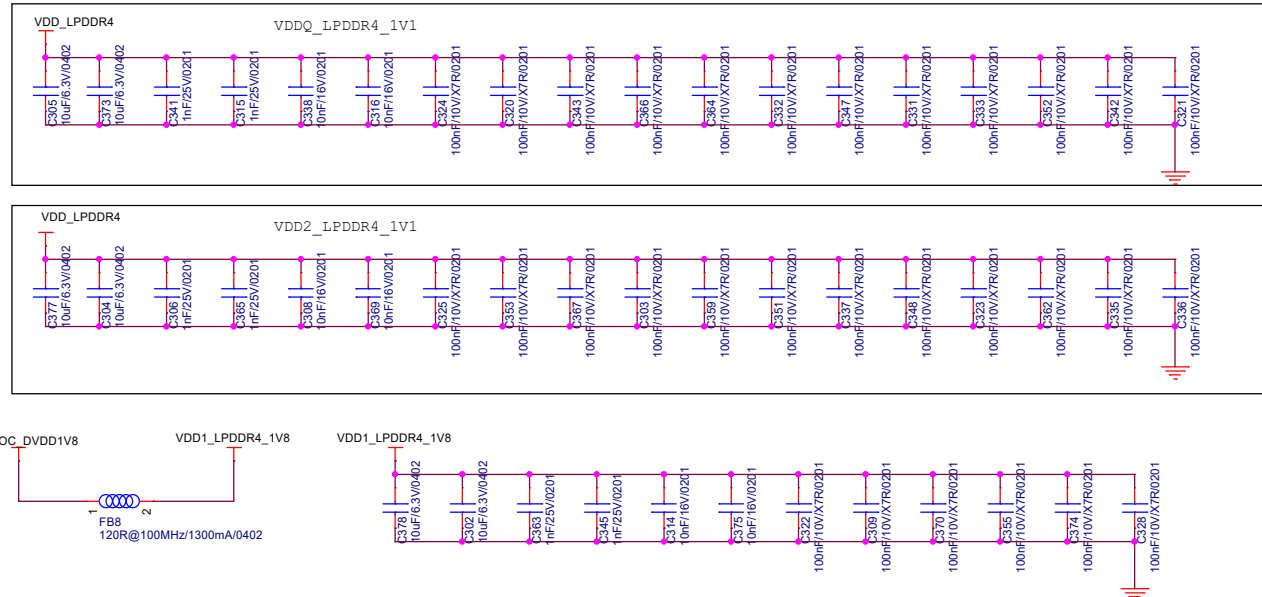
Beijing InHand Networks Technology Co., Ltd.	
Project:	Mo_62A
File:	07. SOC POWER CAPS
Date:	Friday, April 03, 2026
Designed by:	Reviewed by:
Rev:	V1.0
Sheet:	7 of 22

SOC LPDDR4 INTERFACE

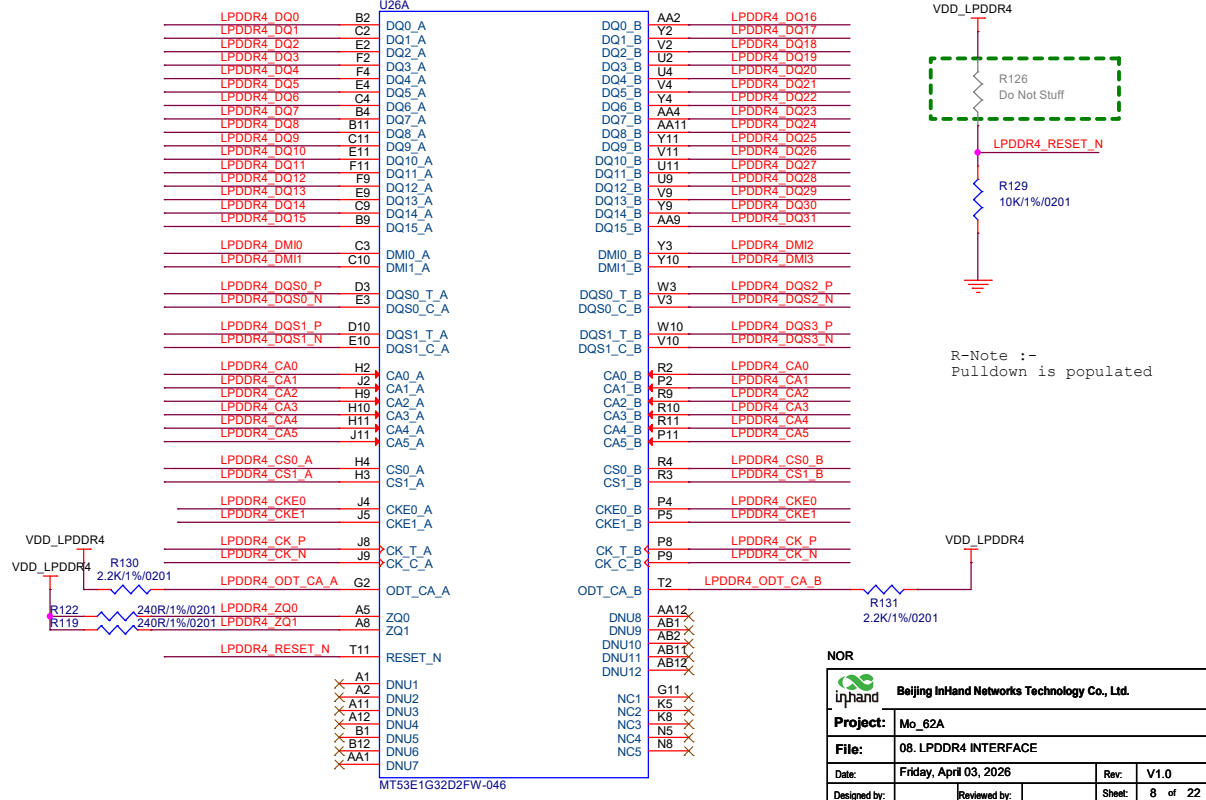
U18G



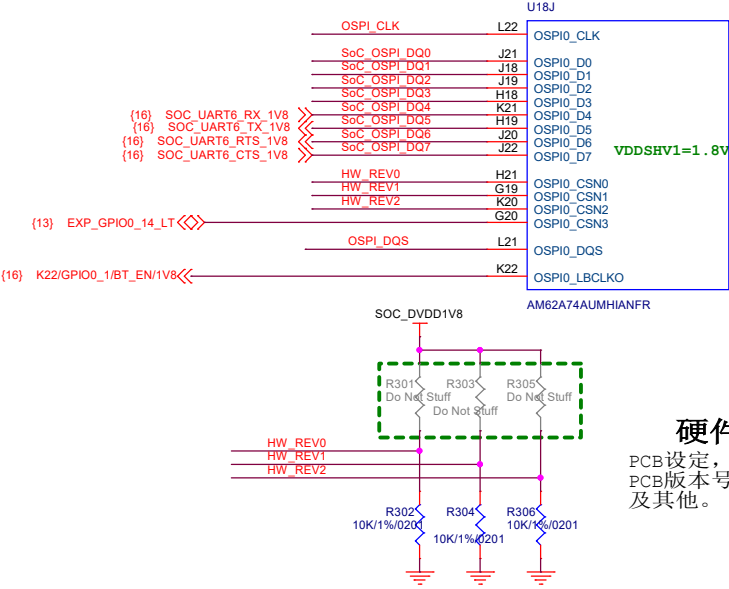
LPDDR4 POWER DECAPS



LPDDR4 DEVICE



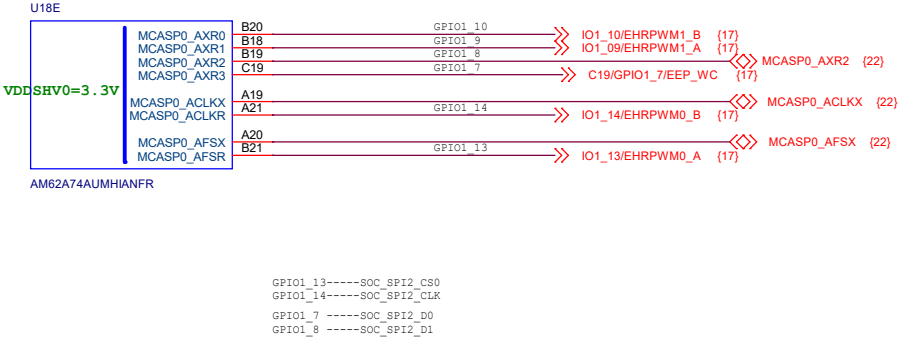
SOC OSPI INTERFACE



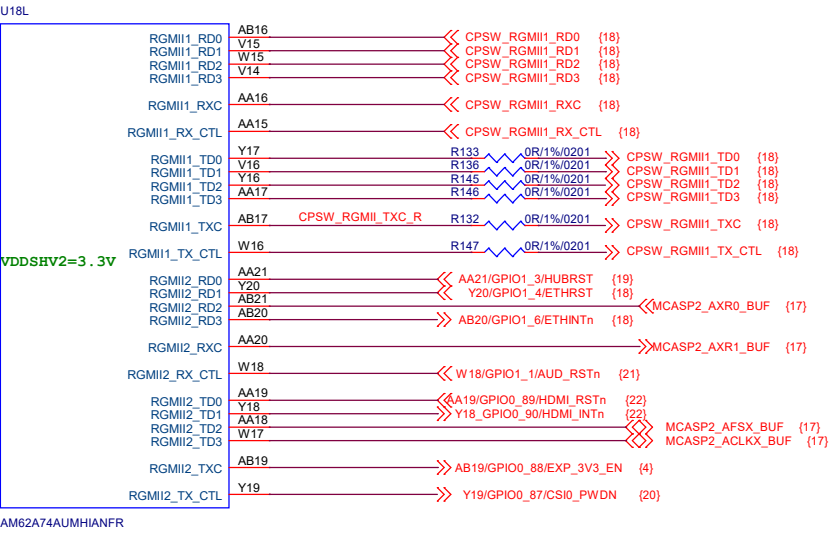
硬件设置

PCB设定，用于设置
PCB版本号、DDR型号、
及其他。

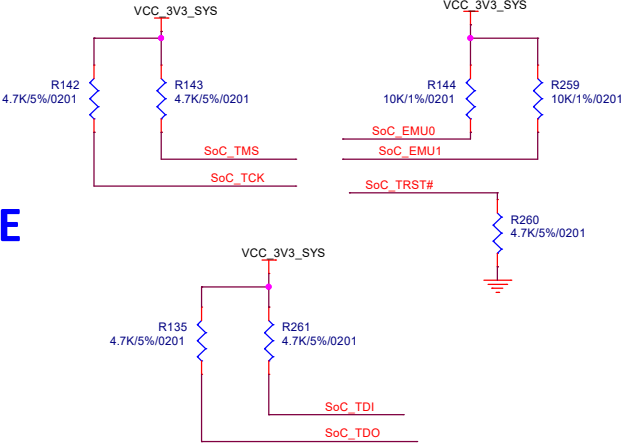
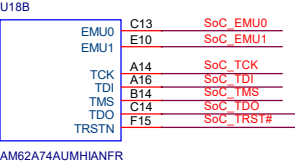
SOC MCASP0 INTERFACE



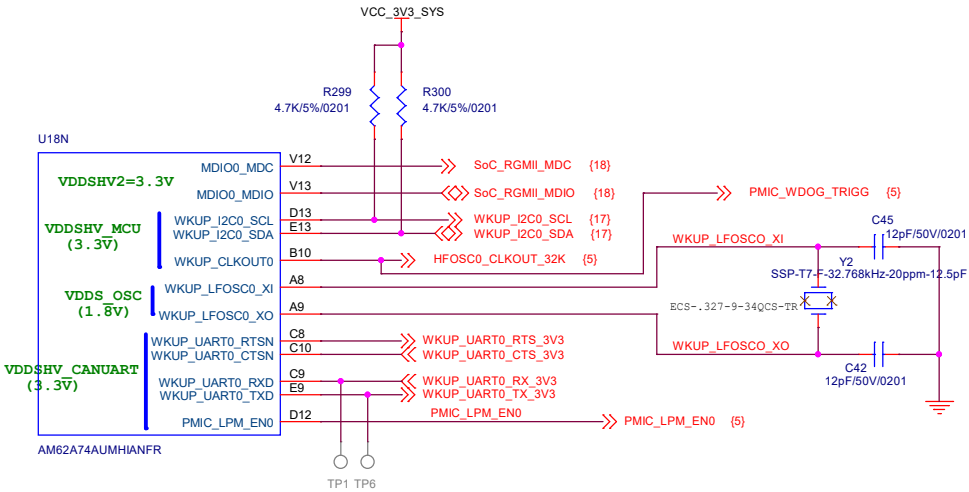
SOC MAC INTERFACE



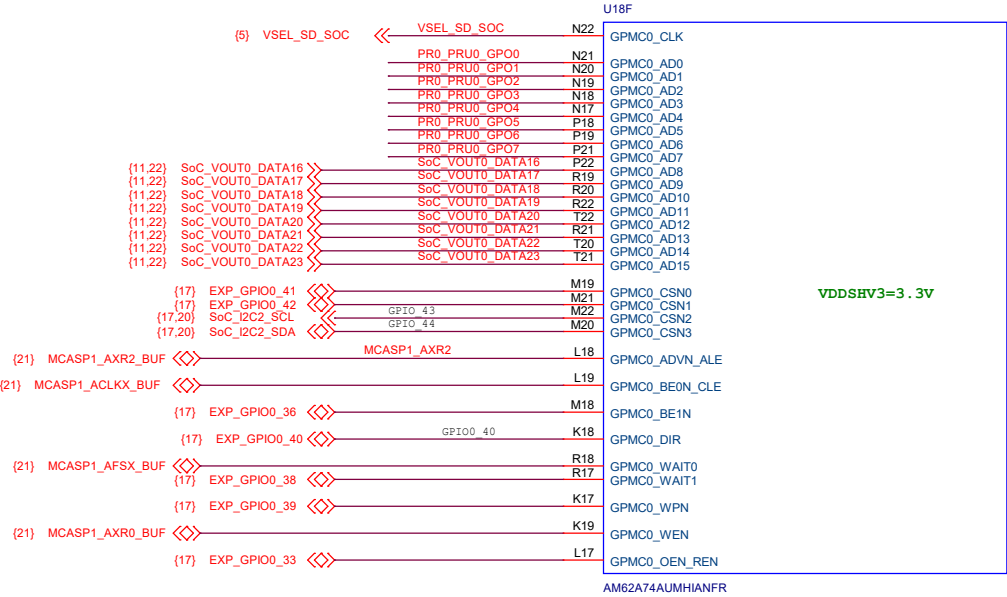
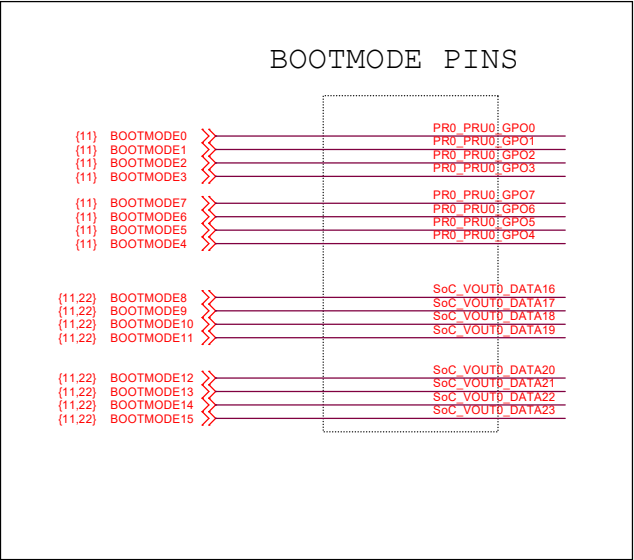
SOC JTAG INTERFACE



SOC WKUP DOMAIN



SOC GPMC INTERFACE



BOOTMODE CONFIGURATION RESISTORS

BOOTMODE7 (10)
BOOTMODE6 (10)
BOOTMODE5 (10)
BOOTMODE4 (10)
BOOTMODE3 (10)
BOOTMODE2 (10)
BOOTMODE1 (10)
BOOTMODE0 (10)

BOOTMODE15 (10,22)
BOOTMODE14 (10,22)
BOOTMODE13 (10,22)
BOOTMODE12 (10,22)
BOOTMODE11 (10,22)
BOOTMODE10 (10,22)
BOOTMODE9 (10,22)
BOOTMODE8 (10,22)

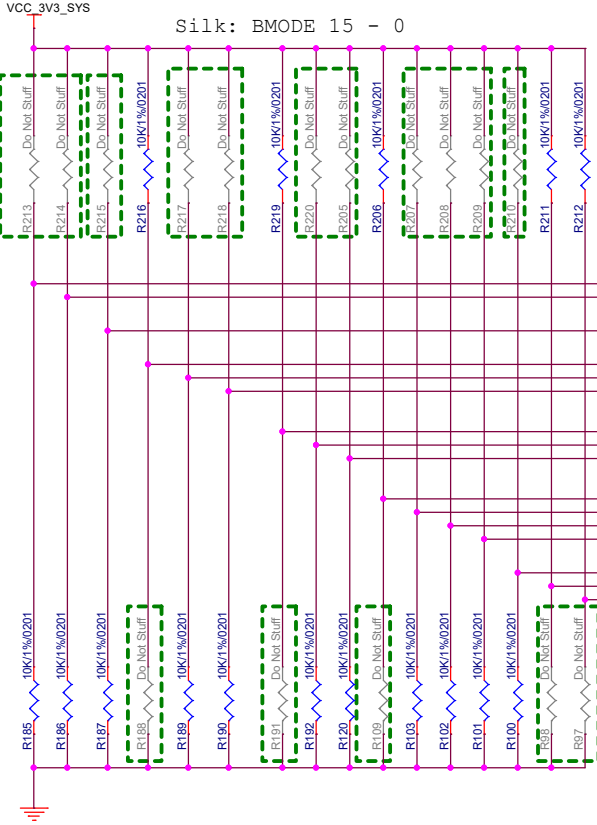
D-Note :-
VCC3V3_XDS TA supply is used to support test automation.
Connect SOC_DVDD3V3 in the custom board design when buffers are not used

D-Note :-
it is not recommended to leave any of the the bootmode pin open including reserved pins

D-Note :-
Dip switch is optional and used on the SK for ease of configuration
A pullup or pulldown resistor can be used to set the BOOTMODE configuration
Provide provision for Pullup and Pulldown resistors for the bootmode pins that have configuration capability

D-Note :-
When dip switches are used on custom board, an external ESD protection may be required if the DIP switches are expected to be configured in an uncontrolled ESD environment

D-Note :-
When DIP switch are used, reduce the resistor values used for the divider to 47K and 470R maintaining the ratio



BOOTMODE15	0	MCU_BOOTMODE[15:14]: Reserved pins.
BOOTMODE14	0	
BOOTMODE13	0	MCU_BOOTMODE[13]: Backup Boot Mode Config, set to '0':RGMI with internal TX delay.
BOOTMODE12	1	MCU_BOOTMODE[12:10]: Backup Boot Mode, set to '100' for Backup boot is Ethernet.
BOOTMODE11	0	
BOOTMODE10	0	
BOOTMODE9	1	MCU_BOOTMODE[9:7]: Primary Boot Mode Config, set to '100'. B9=1: MMC Port 1 (4 bit width). B8=x: Reserved. B7=0: Filesystem mode.
BOOTMODE8	0	
BOOTMODE7	0	
BOOTMODE6	1	
BOOTMODE5	0	
BOOTMODE4	0	MCU_BOOTMODE[6:3]: Primary Boot Mode, set to '1000' for primary boot is MMCSD Boot (SD Card).
BOOTMODE3	0	
BOOTMODE2	0	
BOOTMODE1	1	BOOTMODE[2:0]: PLL Config, set to '011' for 25MHz input frequency.
BOOTMODE0	1	

BOOT MODES SUPPORTED:
Primary Boot: SD CARD
Backup Boot: Ethernet.

BOOT MODES SUPPORTED

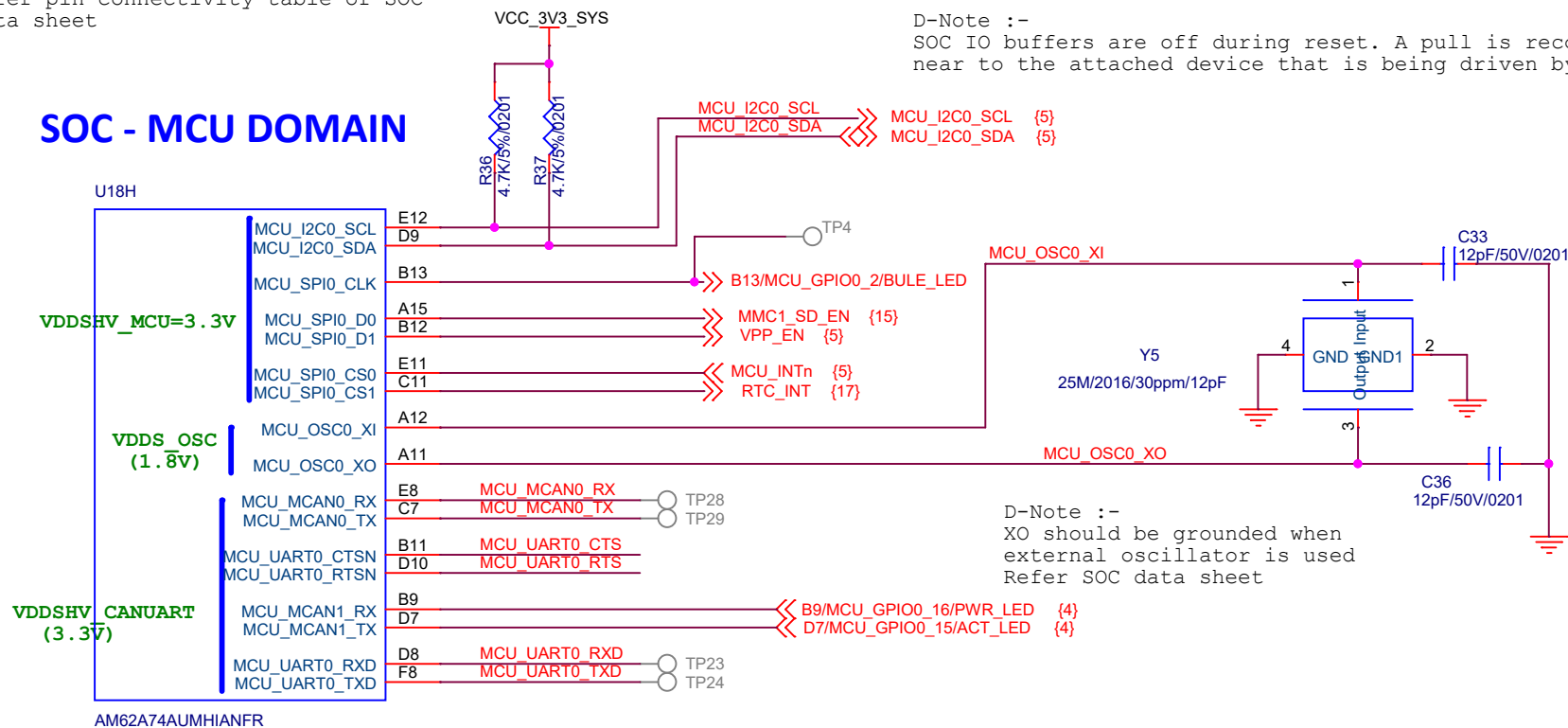
1. OSPI
2. MMC1 - SD CARD
3. UART
4. eMMC
5. ETHERNET
6. USB0 DFU
7. USB0 MS

D-Note :-
A pullup is recommended for Open-drain output type I2C interfaces irrespective of the IO configuration Refer pin connectivity table of SOC data sheet

D-Note :-
Open-drain output type buffer I2C interfaces have slew rate requirement when pulled to 3.3V An RC is recommended for slew rate control. Refer SK-AM62P-LP schematics

D-Note :-
SOC IO buffers are off during reset. A pull is recommended near to the attached device that is being driven by the SOC IO

SOC - MCU DOMAIN



D-Note :-
Connect the 25 MHz crystal directly to the SOC Xi and Xo pins (No Series or parallel resistors are recommended). The internal oscillator implements AGC (Automatic Gain Control) for amplitude control. Match the SOC and the EPHY crystal specs

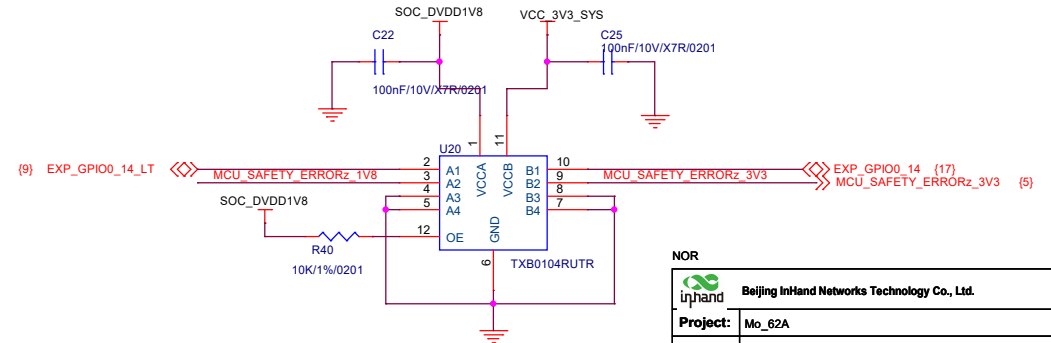
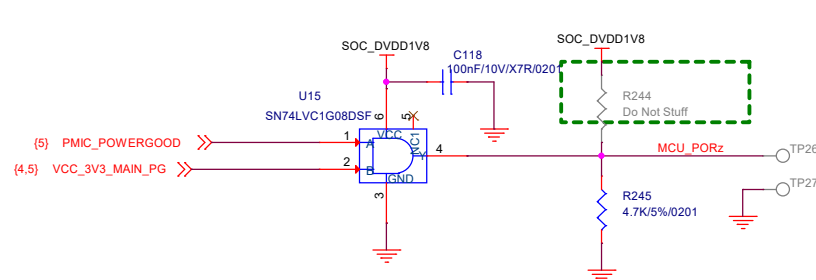
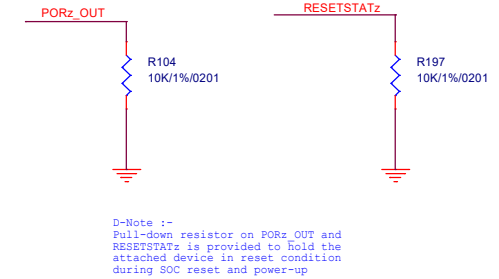
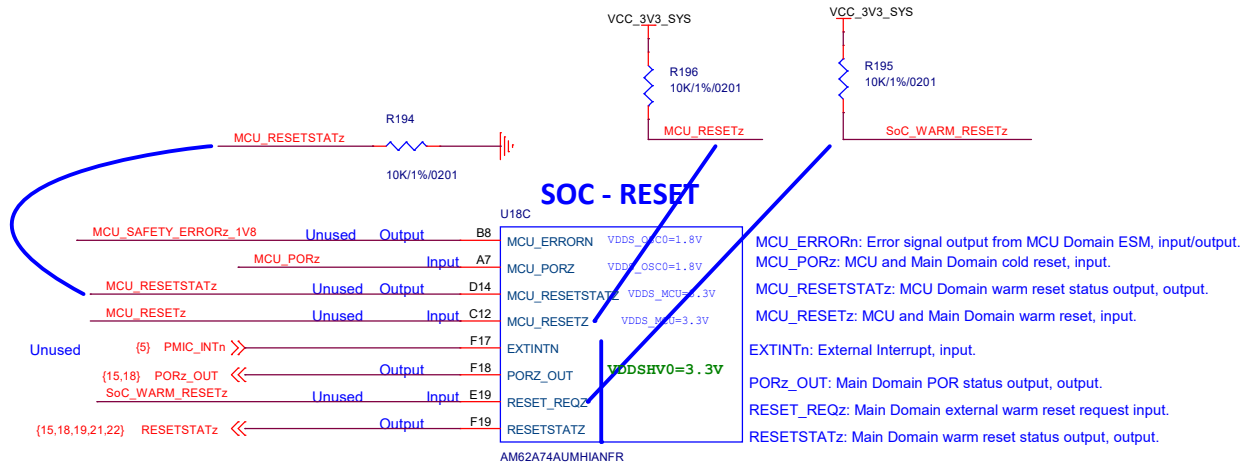
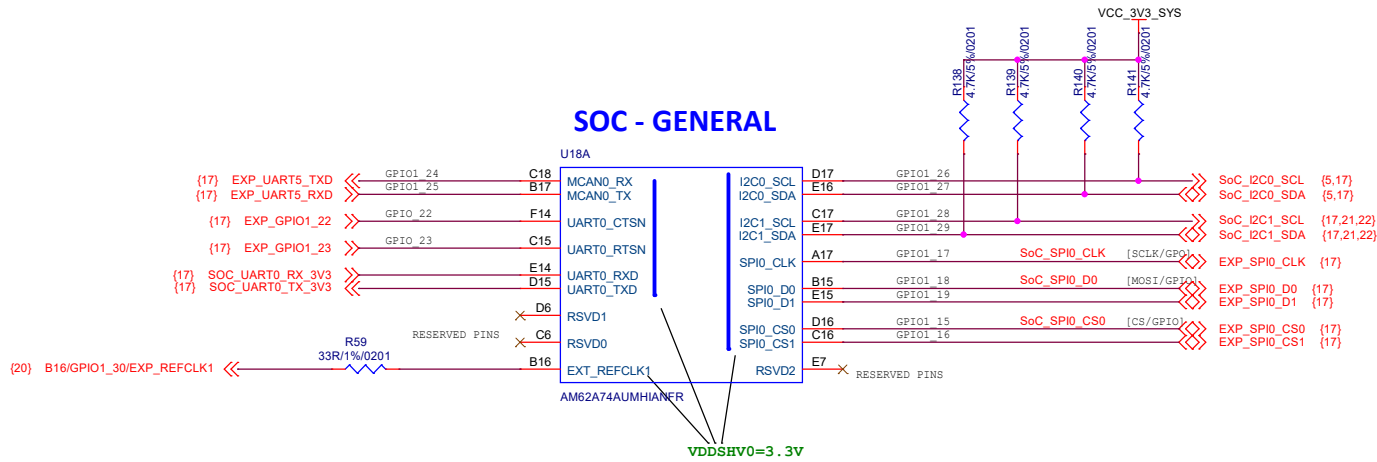
D-Note :-
Refer Applications, Implementation, and Layout section of the data sheet for clock routing guidelines as below:
Clock Routing Guidelines
Oscillator Routing

D-Note :-
No HFOSC0 registers are required to be changed. These registers should remain in their default state. Select the appropriate crystal circuit components that are compliant to the values defined in the MCU_OSC0 Crystal Circuit Requirements table. Read the Load Capacitance and Shunt Capacitance sections to select the appropriate crystal circuit components.

D-Note :-
MCU_OSC0 has been validated only with a 25 MHz clock source, so that is the only frequency supported. The datasheet shows MCU_OSC0 not starting until after the core voltage because there are some cases where the oscillator may not start until VDD_CORE is valid. In most cases it will start as early as VDDS_OSC0, but this may not always be the case. This diagram in the datasheet is showing the maximum start-up time, which must include the case where the delay is based on VDD_CORE being valid.

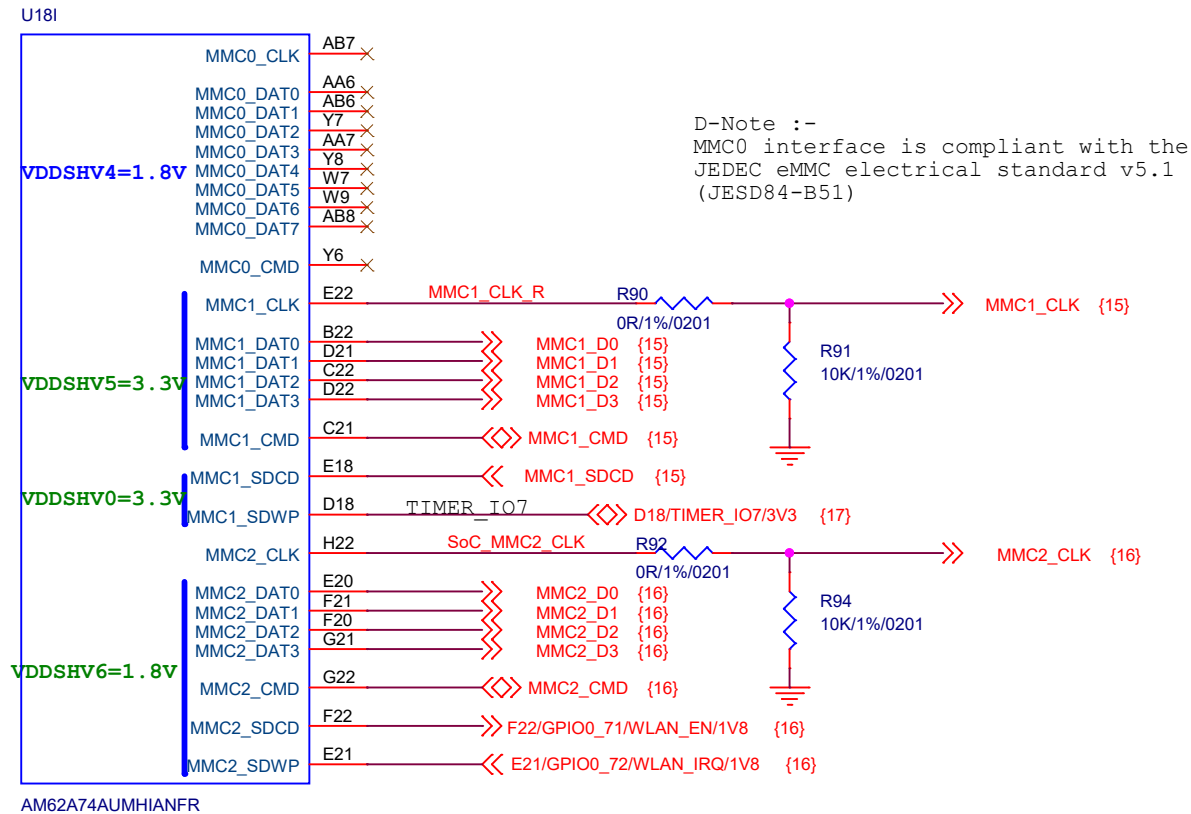
NOR

 Beijing InHand Networks Technology Co., Ltd.	
Project:	Mo_62A
File:	12. MCU HEADER
Date:	Wednesday, April 01, 2026
Designed by:	Reviewed by:
Rev:	V1.0
Sheet:	12 of 22



Beijing InHand Networks Technology Co., Ltd.			
Project:	Mo_62A		
File:	13. OSCILLATOR		
Date:	Friday, April 03, 2026	Rev:	V1.0
Designed by:		Reviewed by:	
Sheet:	13 of 22		

SOC - MMC Interface



NOR

 Beijing InHand Networks Technology Co., Ltd.			
Project:	Mo_62A		
File:	14. MMCx		
Date:	Friday, April 03, 2026		Rev: V1.0
Designed by:		Reviewed by:	
		Sheet:	14 of 22

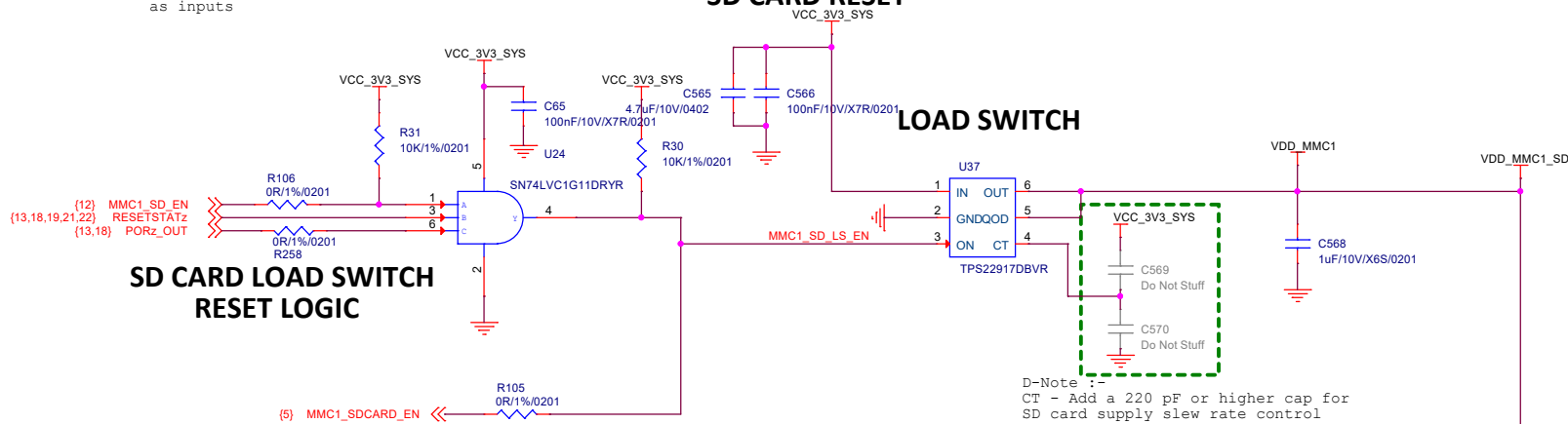
SD CARD INTERFACE

D-Note :-
ANDing logic could be
optimized to 2 input AND gate
Use RESETSTATz and the SOC IO
as inputs

SD CARD RESET

LOAD SWITCH

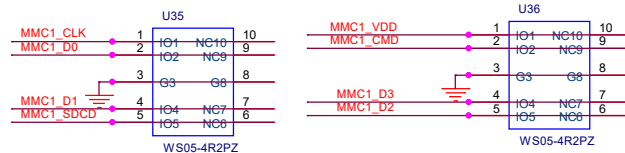
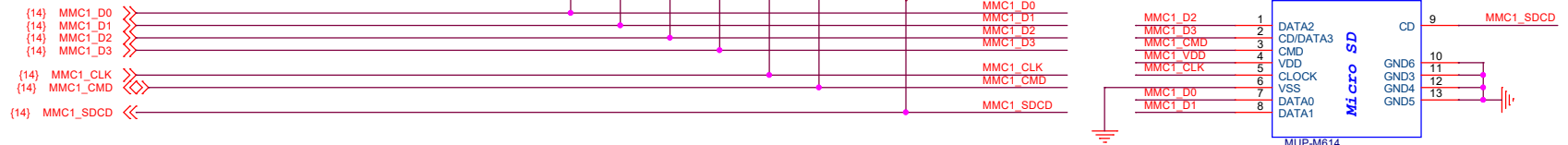
SD CARD LOAD SWITCH RESET LOGIC



R-Note :-
MMC1_CLK pullup is a DNI

D-Note :-
Ensure internal pullups are not configured when 10K
external pullups are used. As a good design practice, a
47K pullup is recommended to ensure the pullup value is
within the SD card specification, when internal pulls
are enabled unexpectedly. This way the resulting pull
resistance will still be within the specified.

D-Note :-
For UHS-I operation, the
pullups are recommended to
be connected to the
3.3V/1.8V switched LDO
output

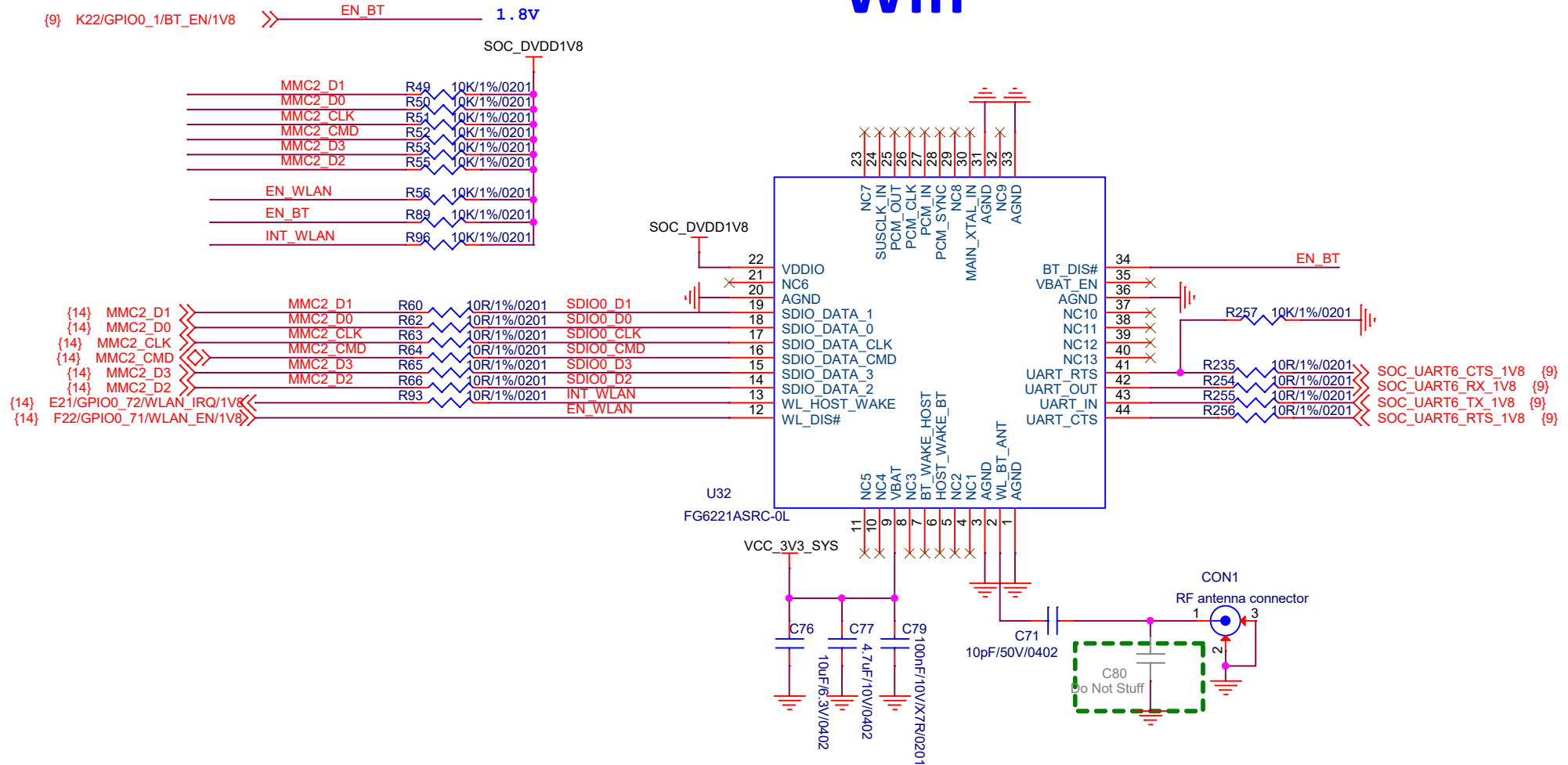


D-Note:-
TO support UHS-I SD Card interface, the pullups are
recommended to be connected to the 3.3V/1.8V
switched LDO output (can be the integrated LDO output)

NOR

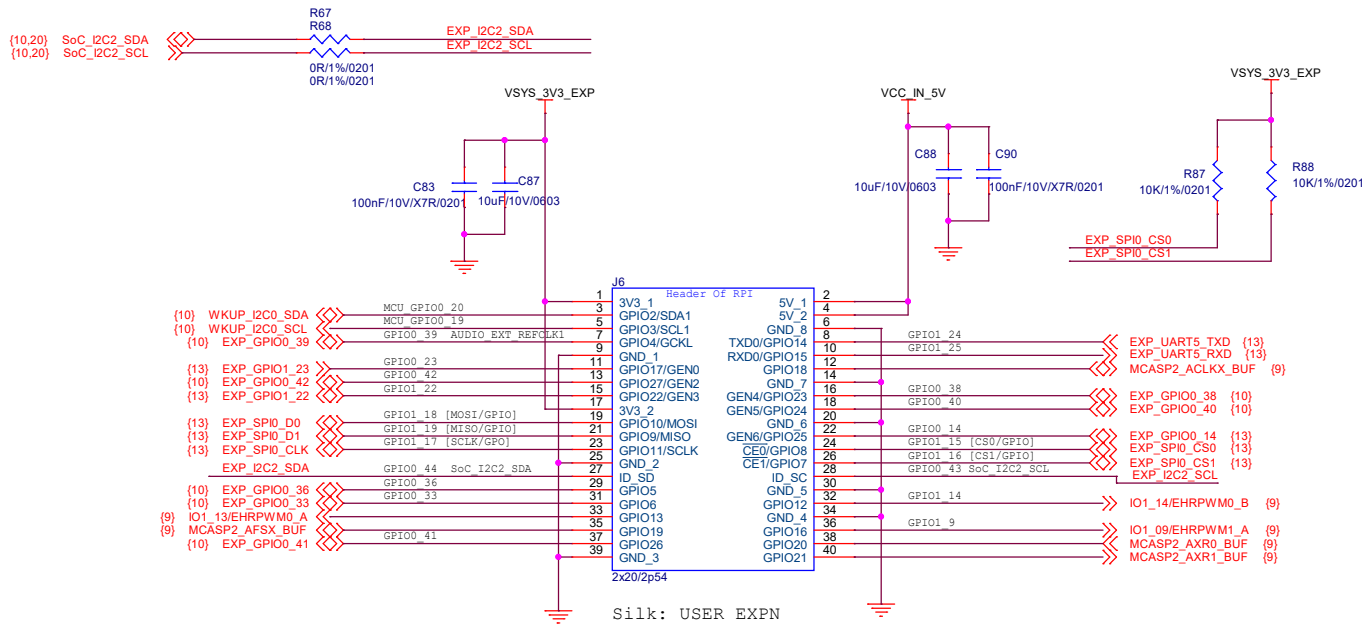
Beijing InHand Networks Technology Co., Ltd.	
Project:	Mo_62A
File:	15. SD CARD INTERFACE
Date:	Wednesday, April 01, 2026
Designed by:	Reviewed by:
Rev:	V1.0
Sheet:	15 of 22

Wifi

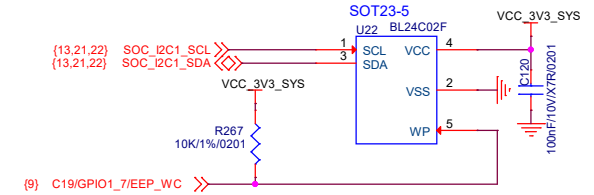


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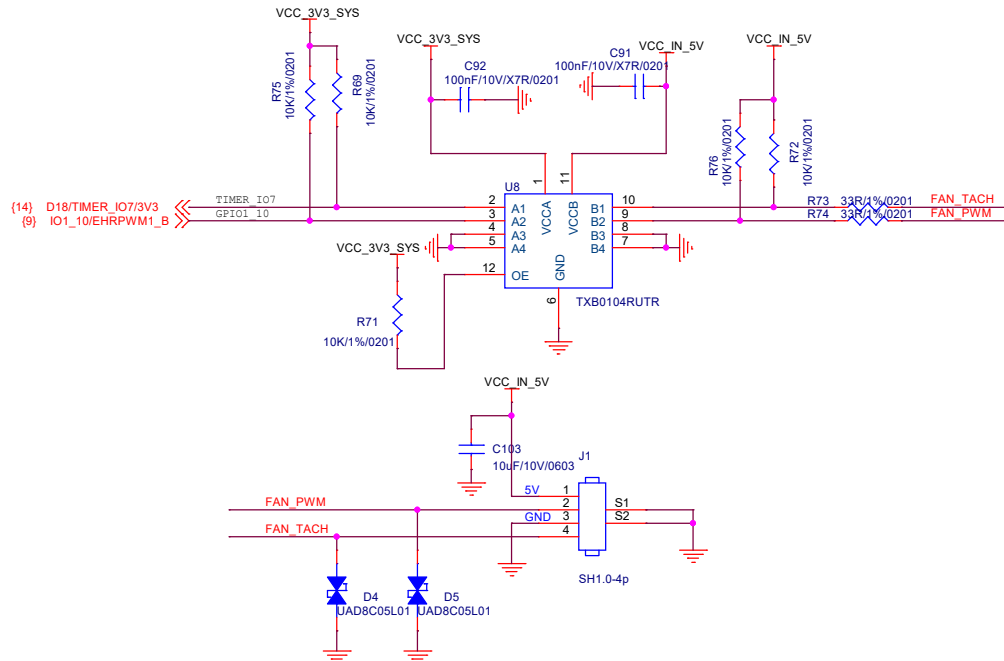
		Beijing InHand Networks Technology Co., Ltd.	
Project:	Mo_62A		
File:	16. WIFI/BLE		
Date:	Friday, April 03, 2026		Rev: V1.0
Designed by:		Reviewed by:	Sheet: 16 of 22



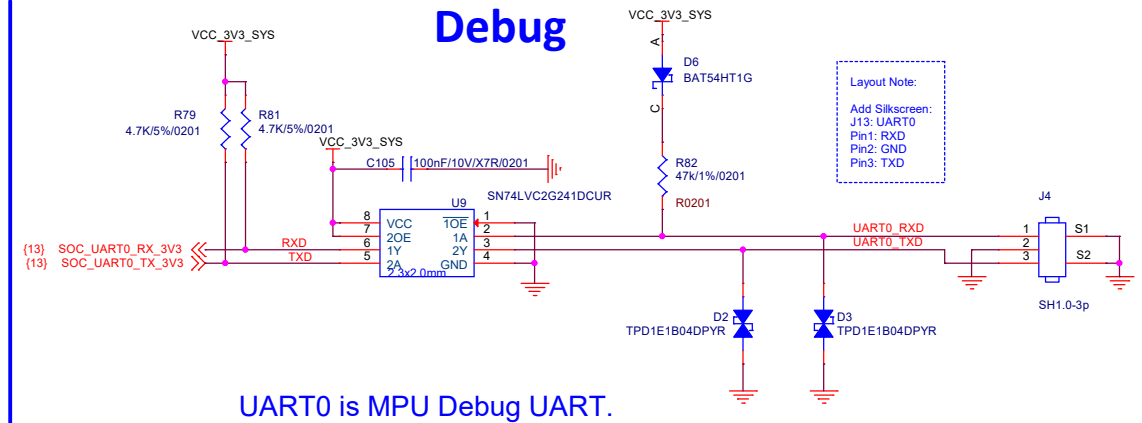
EEPROM



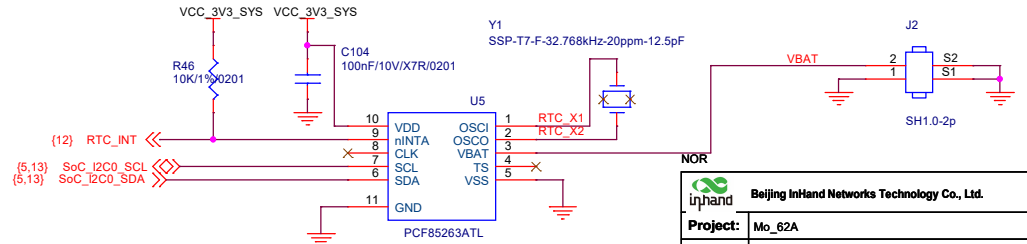
FAN CONNECTOR



Debug

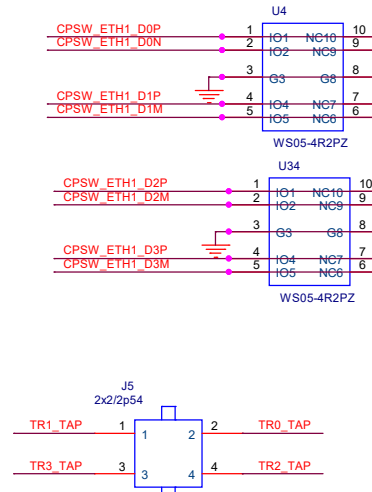
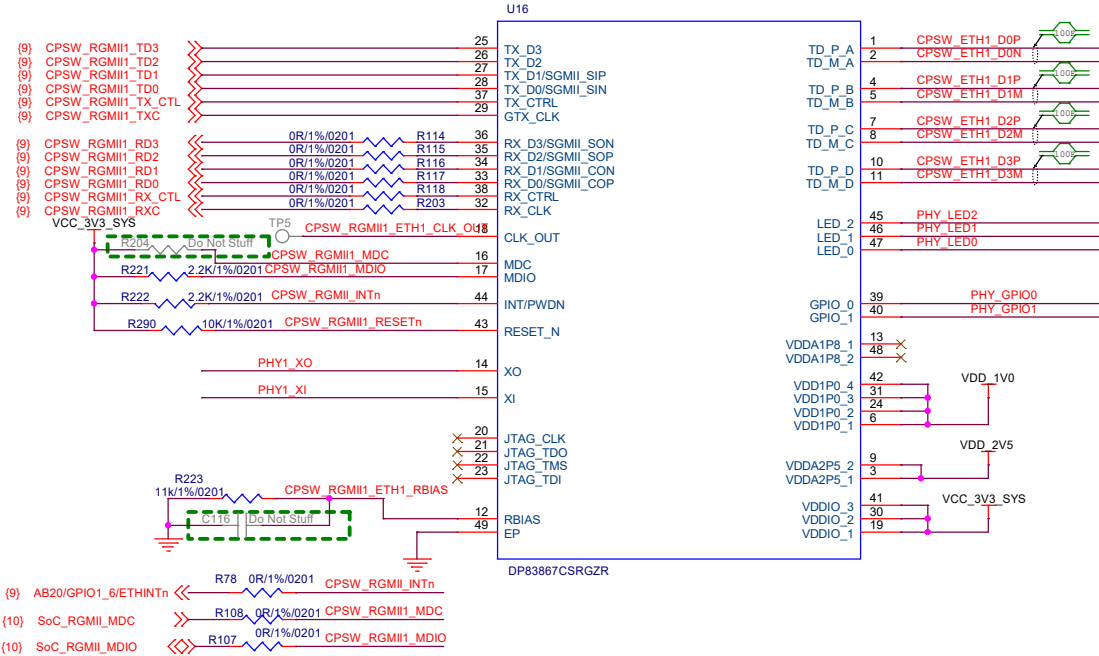
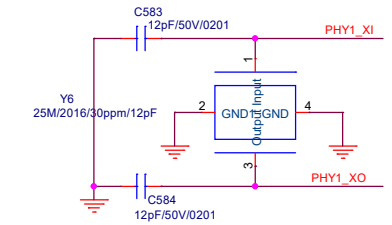
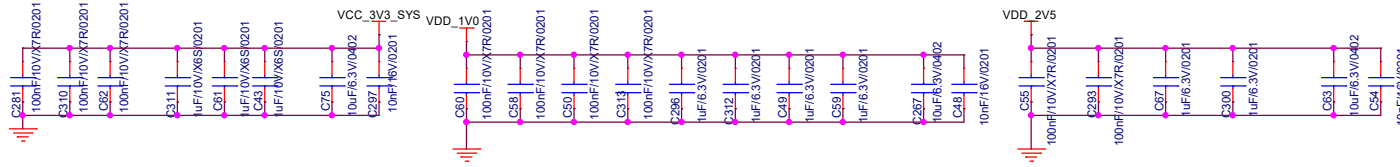


IIC EXT RTC

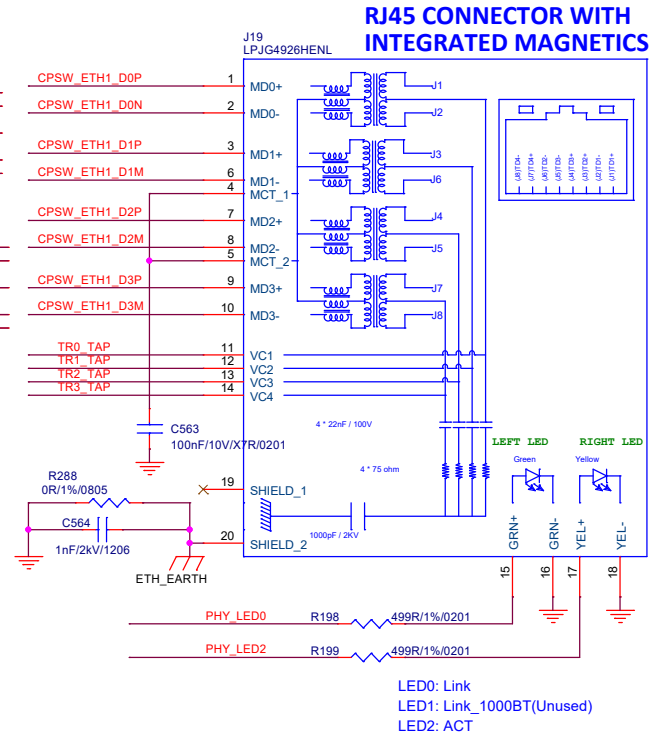


Beijing InHand Networks Technology Co., Ltd.			
Project:	Mo_62A		
File:	17. EXP 40PIN, FAN, RTC, DEBUG		
Date:	Wednesday, April 01, 2026	Rev:	V1.0
Designed by:		Reviewed by:	
Sheet:	17	of	22

CPSW3G RGMII 1 - ETHERNET PHY



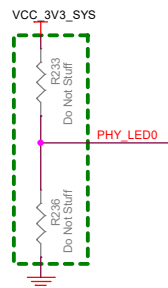
PoE HEADER



PHY ADDRESS = 00000
Auto-negotiation Enabled
10/100/1000 advertised, Auto-MDI-X
Tx Clock Skew = 0ns
Rx Clock Skew = 2ns

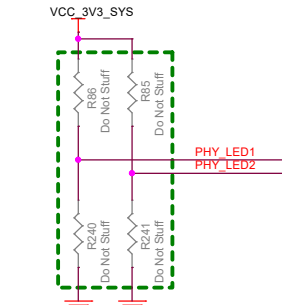
MODE Rh_i Rlo
1 NC NC
2 10k 2.49k
3 5.76k 2.49k
4 2.49k NC

PHY address RXD2, RXD0
set to 0X00

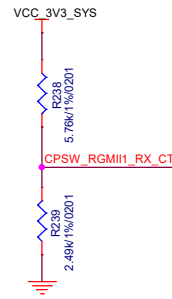


LED_0: MODE 1
Mirror Enable=[0]
For Enabling Mount R233, R236

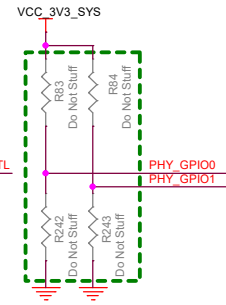
LED1 & LED2: MODE1
RGMII Clock Skew TX[0:2]=[000]
ANEG_SEL(Auto-negotiation)=[0] mean 10/100/1000
PHY_LED[0:2] CANT CONFIGURE IN MODE4



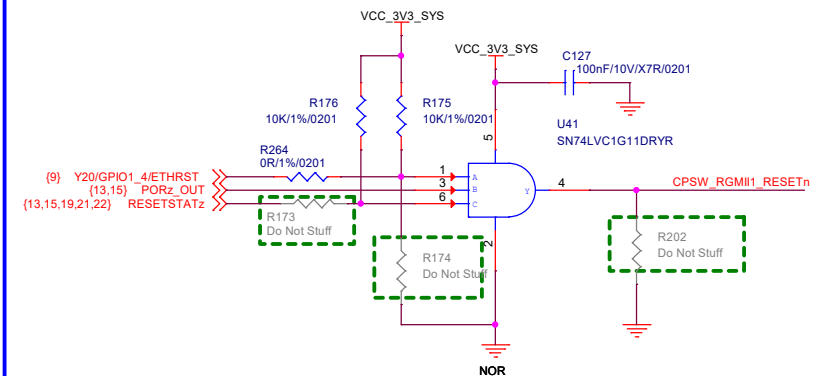
LED1: MODE2 & LED2: MODE1
RGMII Clock Skew TX[2:0]=[000]
ANEG_SEL(Auto-negotiation)=[0]



RX_CTRL:MODE 3
Autoneg enable.
*For Disabling R238=2.49K, R239=N



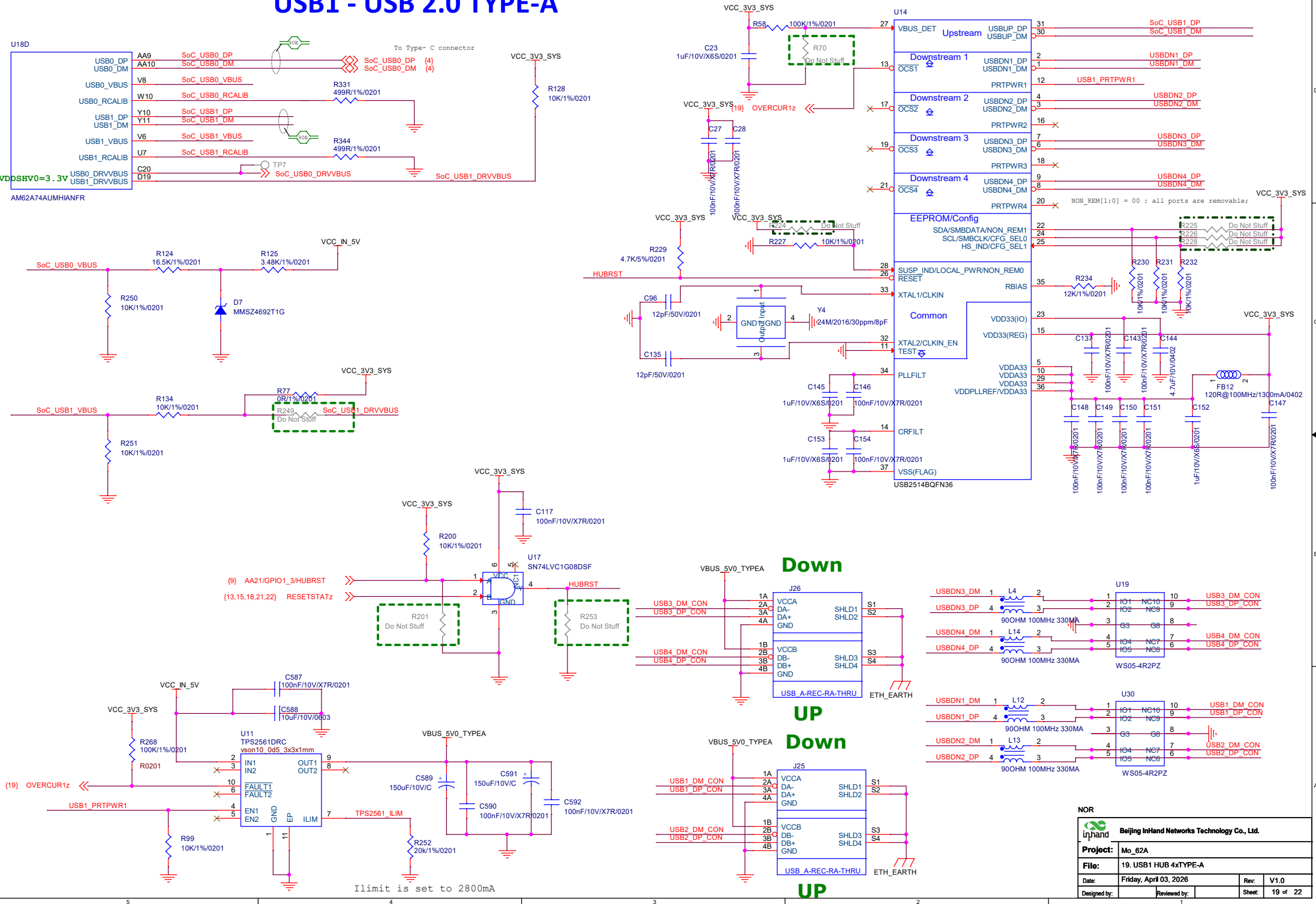
GPIO1 & GPIO0=MODE 1
RGMII Clock Skew RX[0:2]=[000]



NOR

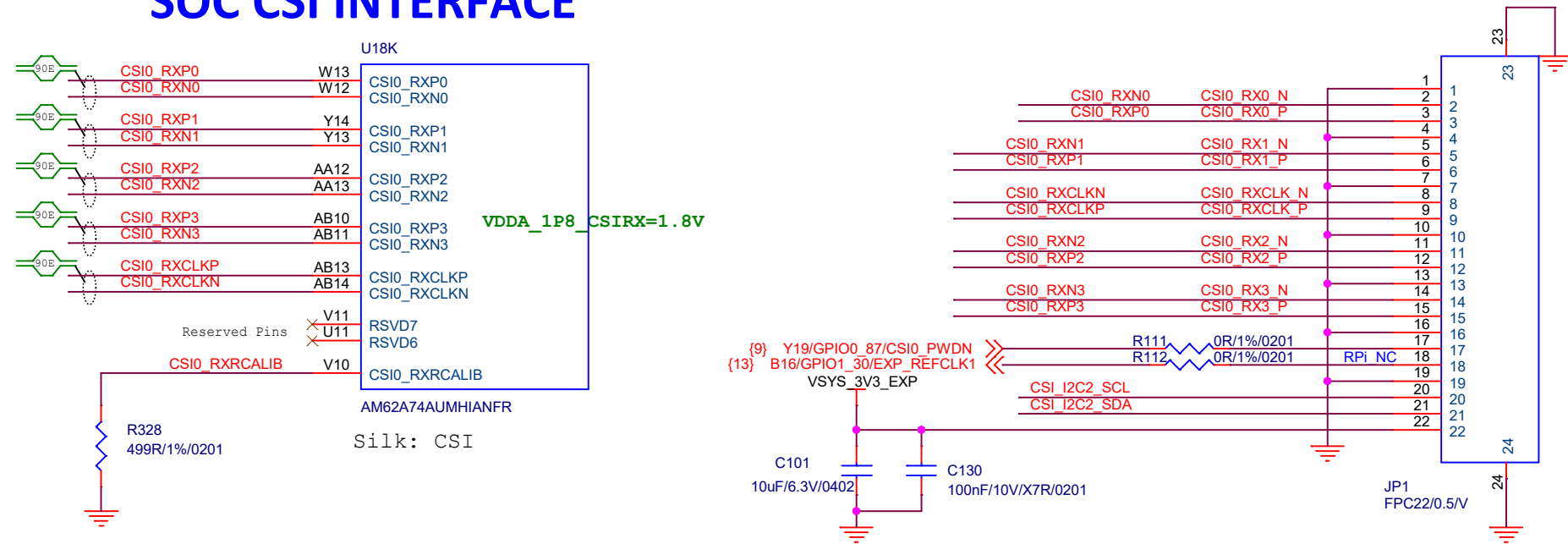
inhand		Beijing InHand Networks Technology Co., Ltd.	
Project:	Mo_62A		
File:	18. ETHERNET		
Date:	Wednesday, April 01, 2026	Rev:	V1.0
Designed by:		Reviewed by:	
		Sheet:	18 of 22

USB1 - USB 2.0 TYPE-A

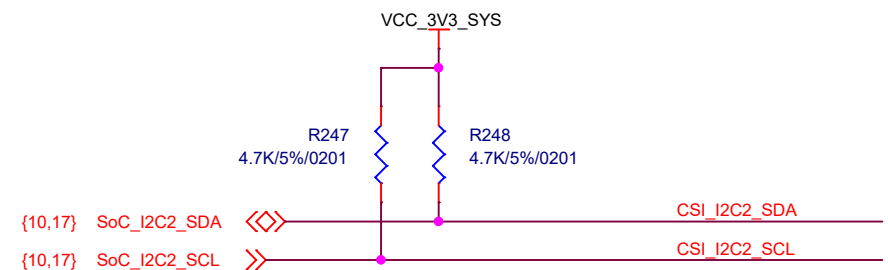


CSI INTERFACE

SOC CSI INTERFACE

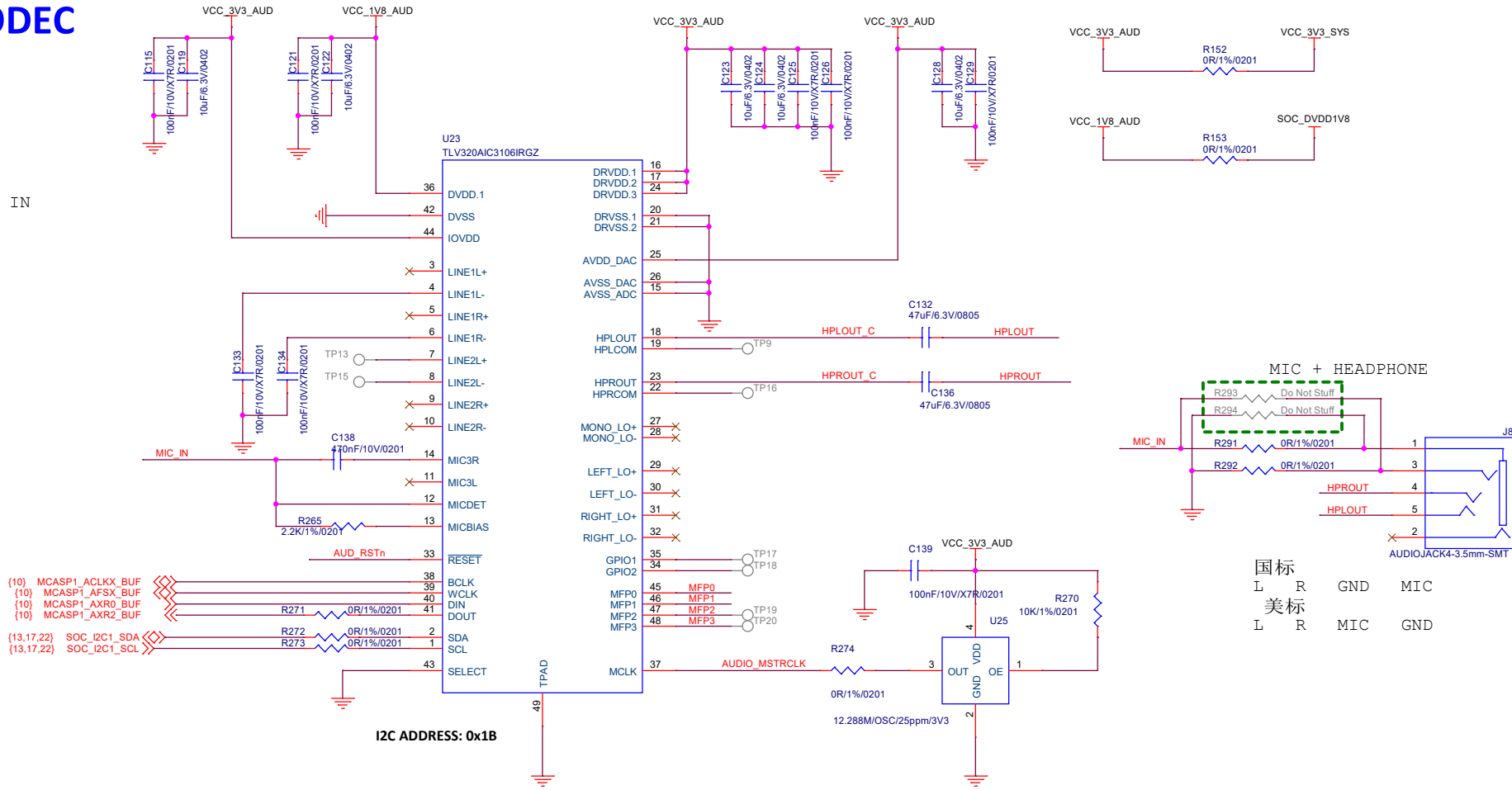


Raspberry Pi Camera Connector x4 Lane



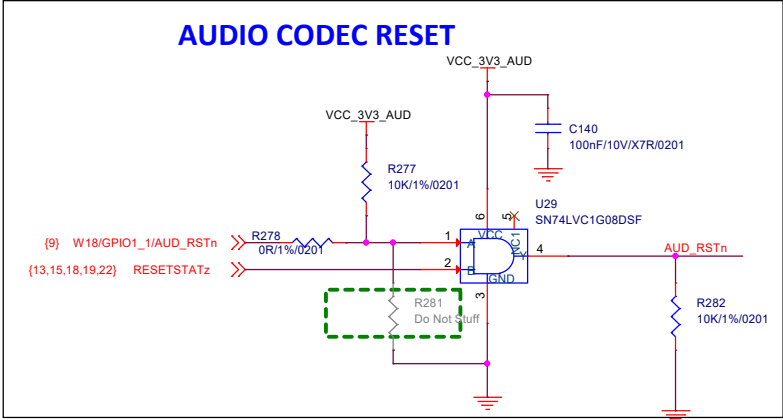
AUDIO CODEC

Silk: LINE IN

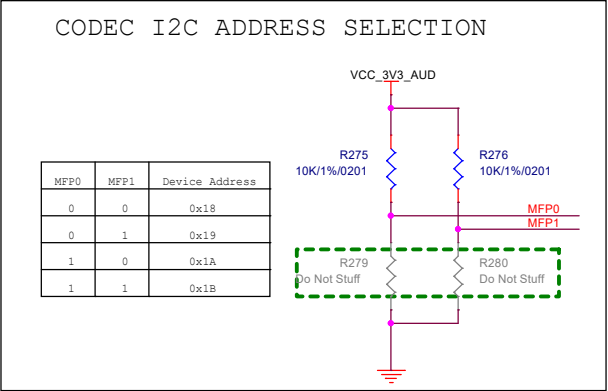


国标
L R GND MIC
美标
L R MIC GND

AUDIO CODEC RESET



CODEC I2C ADDRESS SELECTION



HDMI INTERFACE

R-Note :-
Verify the implementation with
the device manufacturer

D-Note :-
Add an additional decap

U18M

VOUT0_DATA0	U22	SoC VOUT0_DATA0
VOUT0_DATA1	U21	SoC VOUT0_DATA1
VOUT0_DATA2	U20	SoC VOUT0_DATA2
VOUT0_DATA3	U19	SoC VOUT0_DATA3
VOUT0_DATA4	T19	SoC VOUT0_DATA4
VOUT0_DATA5	U18	SoC VOUT0_DATA5
VOUT0_DATA6	V22	SoC VOUT0_DATA6
VOUT0_DATA7	V21	SoC VOUT0_DATA7
VOUT0_DATA8	V19	SoC VOUT0_DATA8
VOUT0_DATA9	V18	SoC VOUT0_DATA9
VOUT0_DATA10	W22	SoC VOUT0_DATA10
VOUT0_DATA11	W21	SoC VOUT0_DATA11
VOUT0_DATA12	W20	SoC VOUT0_DATA12
VOUT0_DATA13	W19	SoC VOUT0_DATA13
VOUT0_DATA14	Y21	SoC VOUT0_DATA14
VOUT0_DATA15	Y22	SoC VOUT0_DATA15
VOUT0_PCLK	AA22	SoC VOUT0_PCLK
VOUT0_DE	U17	SoC VOUT0_DE
VOUT0_VSYNC	V17	SoC VOUT0_VSYNC
VOUT0_HSYNC	T18	SoC VOUT0_HSYNC

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(9) MCASP0_ACLKX	MCASP0_ACLKX_HDMI
(9) MCASP0_AFSX	MCASP0_AFSX_HDMI
(9) MCASP0_AXR2	MCASP0_AXR2_HDMI

SoC VOUT0_DATA0	R154	33R/1%/0201	32	D0
SoC VOUT0_DATA1	R154	33R/1%/0201	31	D1
SoC VOUT0_DATA2	R154	33R/1%/0201	30	D2
SoC VOUT0_DATA3	R154	33R/1%/0201	29	D3
SoC VOUT0_DATA4	R154	33R/1%/0201	28	D4
SoC VOUT0_DATA5	R154	33R/1%/0201	27	D5
SoC VOUT0_DATA6	R154	33R/1%/0201	26	D6
SoC VOUT0_DATA7	R154	33R/1%/0201	25	D7
SoC VOUT0_DATA8	R154	33R/1%/0201	24	D8
SoC VOUT0_DATA9	R154	33R/1%/0201	23	D9
SoC VOUT0_DATA10	R154	33R/1%/0201	22	D10
SoC VOUT0_DATA11	R154	33R/1%/0201	21	D11
SoC VOUT0_DATA12	R154	33R/1%/0201	20	D12
SoC VOUT0_DATA13	R154	33R/1%/0201	19	D13
SoC VOUT0_DATA14	R154	33R/1%/0201	18	D14
SoC VOUT0_DATA15	R154	33R/1%/0201	17	D15
SoC VOUT0_DATA16	R170	33R/1%/0201	16	D16
SoC VOUT0_DATA17	R170	33R/1%/0201	15	D17
SoC VOUT0_DATA18	R170	33R/1%/0201	14	D18
SoC VOUT0_DATA19	R170	33R/1%/0201	13	D19
SoC VOUT0_DATA20	R170	33R/1%/0201	12	D20
SoC VOUT0_DATA21	R170	33R/1%/0201	11	D21
SoC VOUT0_DATA22	R170	33R/1%/0201	10	D22
SoC VOUT0_DATA23	R170	33R/1%/0201	9	D23
SoC VOUT0_HSYNC	R182	33R/1%/0201	34	HSYNC
SoC VOUT0_VSYNC	R182	33R/1%/0201	35	VSYNC
SoC VOUT0_DE	R182	33R/1%/0201	36	DE

R-Note :-
M_CLK is not used

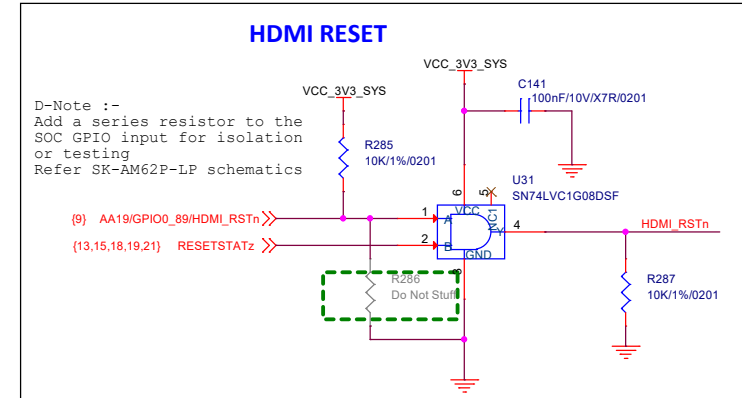
HDMI MSTRCLK	TP22
MCASP0_ACLKX_HDMI	45
MCASP0_AFSX_HDMI	44
WS	44

I2C ADDRESS: 0x3B, 0x3F, 0x62

HDMI TX2+	U61	ESD73034D	1	IO1	NC10
HDMI TX2-	U61	ESD73034D	2	IO2	NC9
HDMI TX1+	U61	ESD73034D	3	G3	G8
HDMI TX1-	U61	ESD73034D	4	IO4	NC7
HDMI TX0+	U60	ESD73034D	1	IO1	NC10
HDMI TX0-	U60	ESD73034D	2	IO2	NC9
HDMI CLK+	U60	ESD73034D	3	G3	G8
HDMI CLK-	U60	ESD73034D	4	IO4	NC7
HDMI I2C_SCL	U59	WS05-4R2PZ	1	IO1	NC10
HDMI I2C_SDA	U59	WS05-4R2PZ	2	IO2	NC9
HDMI HPD	U59	WS05-4R2PZ	3	G3	G8
HDMI CEC	U59	WS05-4R2PZ	4	IO4	NC7

VCC_IN_5V	D9	B5817WS
HDMI HPD	D10	UAD8C05L01

Micro_HDMI_TypeD_19P
Micro_HDMI_TypeD_19P



Beijing InHand Networks Technology Co., Ltd.			
Project:	Mo_62A	Rev:	V1.0
File:	22. HDMI INTERFACE	Sheet:	22 of 22
Date:	Friday, April 03, 2026	Reviewed by:	
Designed by:		Reviewed by:	